



Project: Serialization

(Due on on Léa)

Overview

For the project, you will be adding *serialization* to a selection of objects and data structures we studied in class. Serialization is the process of translating an object of data structure residing in memory into a series of bytes that can be stored to file or communicated over a network. The inverse process, converting the data structure from bytes back into memory, is called deserialization.

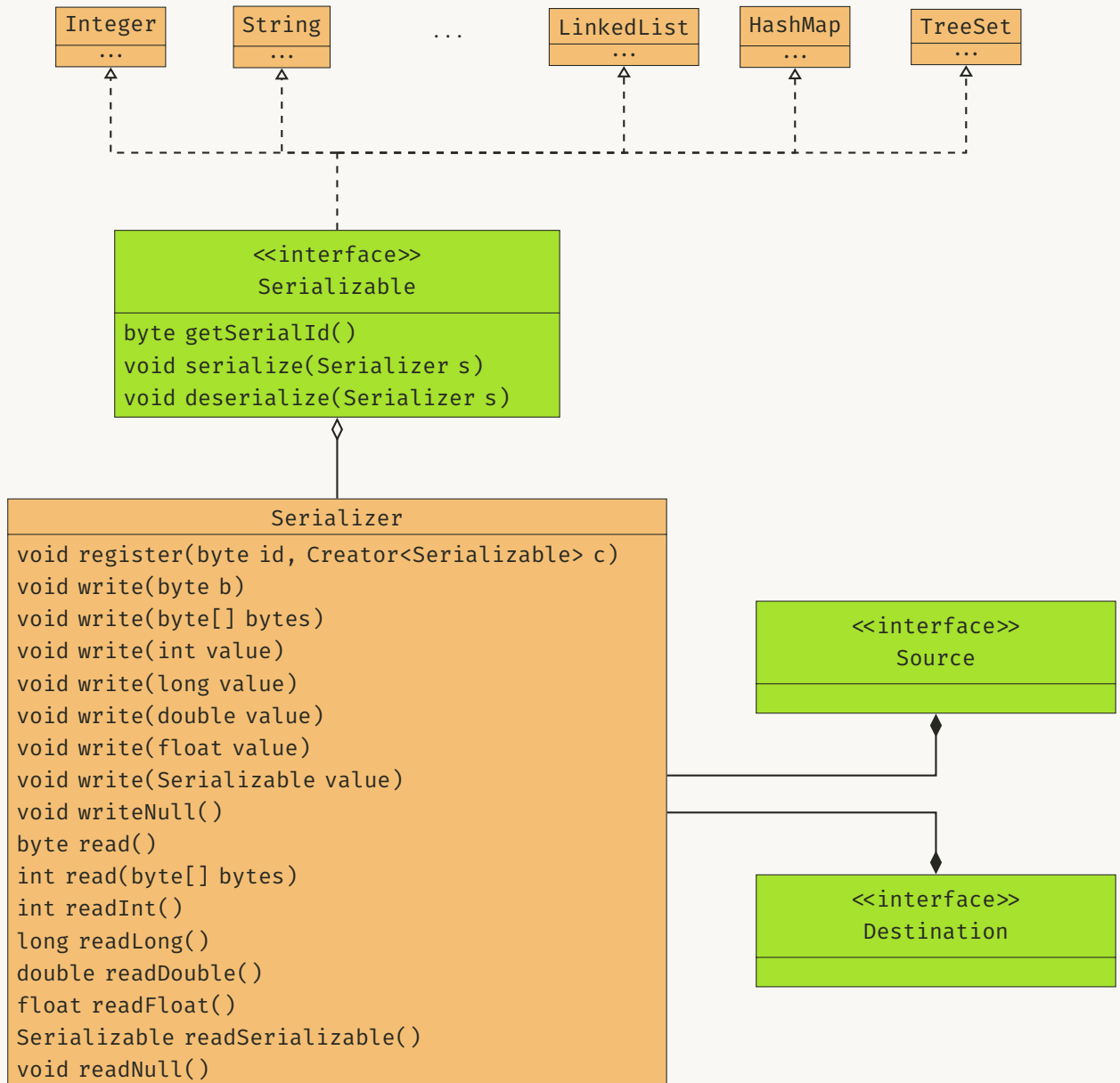
You will write serialization and deserialization operations for some of the classes in this course.

Example. Here is the serial representation of the string “hello world”.

0x02	0x0000000b	0x68	0x65	0x6c	0x6c	0x6f	0x20	0x77	0x6f	0x72	0x6c	0x64
id	length	h	e	l	l	o		w	o	r	l	d

1 Project Design

The design of the project is given in the following UML class diagram. You will be writing classes that implement the `Serializable` interface and will update the `Serializer` class. You will need to use these classes to demonstrate that your serialization works.



1.1 Serializable interface

Classes to be serialized will need to implement the `Serializable` API. It has the following operations:

Signature	<code>byte getSerialId()</code>
Pre-conditions	
Post-conditions	
Returns	Return the serial identifier set for objects of this class.

Signature	<code>void serialize(Serializer s)</code>
Pre-conditions	A serialization header has been written to the output (see below).
Post-conditions	A binary representation of the object has been written to the output. This representation is sufficient to create an identical copy when deserializing.
Returns	

Signature	<code>void deserialize(Serializer s)</code>
Pre-conditions	The serialization header has been read from the input, and validated (see below).
Post-conditions	The binary representation has been read from the input and the object state has initialized as an identical copy of the serialized version.
Returns	

1.2 Serializer class

While each class is in charge of generating their own serial representation, a `Serializer` will manage the interaction with the input source and output destination and support the reading and writing of primitives. All serializable classes will be passed a `serializer` to write/read their representations to.

While the complete `Serializer` API is available in the code documentation, here some of it's important methods:

Writing methods. Here is a description of three of the writing methods. Each primitive has it's own `write` method.

Signature	<code>void write(byte[] bytes)</code>
Pre-conditions	
Post-conditions	The bytes are written to the output destination.
Returns	

Signature	<code>void write(int value)</code>
Pre-conditions	
Post-conditions	The integer value is written to the output destination.
Returns	

Signature	<code>void write(Serializable s)</code>
Pre-conditions	
Post-conditions	A record header is written to the output destination, followed by the serial representation.
Returns	

Reading methods. Here is a description of three of the reading methods. Each primitive has it's own read method.

Signature	<code>Serializable readSerializable()</code>
Pre-conditions	
Post-conditions	The record header is read and validated, the object is constructed and the serial representation is used to initialize it.
Returns	The deserialized object.

Signature	<code>int read(byte[] bytes)</code>
Pre-conditions	
Post-conditions	The array is filled with the next bytes read from the input source.
Returns	The number of bytes read.

Signature	<code>int readInt()</code>
Pre-conditions	
Post-conditions	Read the next integer from the input source.
Returns	The integer read.

1.3 The record header

Each serialized object will have a *record header* that is written to the output destination during serialization and then read from the input source during deserialization:

header ₁	body ₁	header ₂	body ₂	header ₃	body ₃	...
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So the output is a series of records separated by headers. The header will contain the object's class *serial ID* and in Section 3 you will add more.

1.4 Serial IDs

Each `Serializable` class must have a unique ID given to it. In this project, we will use byte IDs, but in practice larger numbers are used¹. This ID serves two important purposes in deserialization:

1. It is a *sanity* check, which means that when deserializing data, we can check that the thing we're about to deserialize is what we expect to be deserializing. Recall that binary data has many interpretations. This ID will help us make sure we have what we want.
2. We will use the serial ID to create an instance of the correct class. We need this instance to call the `deserialize()` method and initialize the object from its serial representation.

The second item means we need some mechanism to create objects. The way I have chosen to set this up is to register a default constructor (a constructor with no arguments) for each serializable we intend our `Serializer` to serialize. For example,

```
Serializer serializer = new Serializer(..);
serializer.register(Integer.SERIAL_ID, Integer::new);
serializer.register(String.SERIAL_ID, String::new);
serializer.register(Box.SERIAL_ID, Box::new);
```

will setup my `Serializer` to serialize and deserialize `Integer`, `String` and `Box<T>` objects. Note that primitives do not need to be registered.

2 Serializable Classes

Here is a list of classes to make `Serializable`. Some I've done already as examples for you to study in order understand how to design serial representations for the others. All package names are relative to `ca.qc.johnabbott.cs406` in the starter project.

¹Java's provided serialization uses longs.

Class name	Notes
<code>serialization.util.Integer</code>	Provided. Serializable wrapper class for integers.
<code>serialization.util.String</code>	Provided. Serializable wrapper class for <code>String</code> . Serial representation is an int and a char[].
<code>serialization.util.Box</code>	Provided. A generic class that uses the fact that its content is serializable.
<code>serialization.util.Date</code>	Serializable wrapper class for <code>java.util.Date</code> . Use a date's numeric representation.
<code>IPAddress</code>	The representation of an IP address as a byte[] from earlier in class.
<code>Grade</code>	A simple data classes using fields you can already serialize.
<code>collections.Either</code>	A generic interface, whose implementations store one of two possible values of two distinct types. To ensure that all implementing classes are serializable, add the line <code>extends Serializable</code> to the interface declaration and implement the required methods.
<code>collections.list.LinkedList</code>	Serialization should store all the necessary data to reconstruct the linked list. Deserialize without calls to <code>add(elem)</code> or <code>add(pos, elem)</code> .
<code>collections.map.HashMap</code>	Serialization should store all the necessary data to reconstruct the hash map, including the number of buckets and the maximum load factor. Deserialize without calls to <code>put(key, value)</code> or even recomputing the bucket. Hint: store and load all colliding records together.
<code>collections.set.TreeSet</code>	Serialization should store and load not only the set data, but also the tree structure they are in. Deserialize without calls to <code>add(elem)</code> . Hint: use one of the tree traversal algorithms to write and subsequently reconstruct the tree. You will add a second generic bound to your class declaration. See https://docs.oracle.com/javase/tutorial/java/generics/bounded.html .

For each class, describe its serial representation in the class comment at the top of the .java file.

Advice. Although your submission must meet the above restrictions, i.e.: not calling operations like `add( .. )` and `put( .. )`, maybe your first attempt at implementation can call them to see if you're on the right track.

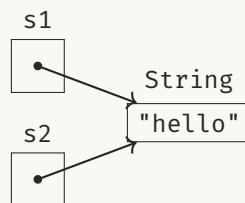
3 Optimizing References

Recall from the course that the building blocks of data structures (and classes) are references, and that we can use references to create object aliases. We want a way to represent aliases, both to ensure identical original and deserialized data and to save on the size of the output, by removing duplicates.

We will implement this by adding information to the record header. Each header will now contain a *flag* to indicate if it the body is a serial representation or an alias. If it's an alias, then the body is simply an integer indicating the byte index of the record containing the object. In the `Serializer` class, I've added constants:

```
public static final byte ALIAS_MARKER = (byte) 0xff;
public static final byte ORIGINAL_MARKER = 0x00;
```

Example. For the following,



serializing `s1` then `s2`, we would get:

0x02	0x00	5	h	e	l	l	o	0x02	0xff	0x0000
id	flag							id	flag	index

The last field, the byte index, would depend on where in the output destination `s1` had been serialized.

3.1 Hints

I suggest you add two maps to the `Serializer` class. The first is a `Map<Serializable, Integer>` that will store the byte index of each object that has been serialized to the output destination. The second is a `Map<Integer, Serializable>` that will store the deserialized objects by their byte index in the input source.

The first map is a `IdentityHashMap`², the second is the usual `HashMap`. The difference between these two classes is that, when searching by key, the identity version uses `=` instead of `.equals()`. When serializing, you will keep track of the things you've previously serialized, and when you are asked to serialize something you've already done you can perform the optimization.

Requirements

Your program *must* meet the following requirements:

1. Your program should be clear and well commented. It must follow the “420-406 Style Guidelines” (on Léa).
2. Start from the provided classes.
3. Implement `Serializable` for the remaining classes in Section 2. Document the format that you use in code comments.
4. For `LinkedList<T>` deserialization, do not call `add(elem)` or `add(pos, elem)`.
5. For `HashMap<K, V>` do not call `put(key, value)`.
6. For `TreeSet<T>` do not call `add(elem)`.
7. Update `Serializer` to optimize references as described in Section 3.
8. Submit using Git by following the instructions (on Léa).

²Here you will need to use the Java libraries, since we did not implement the `IdentityHashMap`.